Course Type	Course Code	Name of Course	L	T	P	Credit
DE	NGPD510/GPD508	Seismological Data Analysis	3	0	0	3

Course Objective

This course provides hands-on experience with various seismological analysis techniques, including Linux, Bash scripting, Fortran, MATLAB, and Python. One key outcome is a deeper understanding of these methods, the underlying theory, and their practical applications. Another outcome is the development of skills in managing digital data—acquiring, editing, manipulating, and filtering it effectively. The aim is to equip you with a working knowledge of common techniques and programs used in seismology, which will be valuable whether you conduct research using these methods or utilize analysis results such as earthquake locations, receiver functions, or tomographic models. Additionally, this course will enhance your proficiency with computational tools like MATLAB, Linux, Bash scripting, and Python.

Learning Outcomes

This course offers students hands-on experience with a range of seismology tools, utilizing Linux, Python, and MATLAB. Additionally, they will have the chance to explore widely used seismological tools for effective data analysis.

	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Introduction to Unix/Linux, AWK, Bash Shell Scripting	04	Learning Linux/Unix to handle high volumes of seismological data.
2.	Types of Data, Seismological Data request tools, Standing Order of Data	03	Students will learn to request earthquake data.
3.	Basics of Python, Introduction to ObsPy to access data from data centers	03	Tool to request and process earthquake data in Python.
4.	Seismic data formats (e.g., SEED, SEGY, etc.), Processing digital seismogram, RESP files, dataless SEED files	04	Various seismic data formats and instrument response files.
5.	Intro to Seismic Analysis Code (SAC), Identifying Phases, Picking arrivals, Plotting and printing SAC files, frequency and filtering, Using SAC externally	04	Hands-on experience for identifying and picking seismic phases.
6.	Removal of instrument response, Seismogram rotation, cross-correlation, stacking various phases	03	Rotating a seismogram and instrument response removal.

7.	Generic Mapping Tools (GMT), Plotting 2-D maps of geophysical information, Using Colormaps to convey 3-D information	04	Plotting geophysical information on maps using GMT script.
8.	Introduction to MatLab, Spectral Analysis in MatLab	03	Using MatLab as a tool to perform spectral analysis.
9.	Normal modes of Earth, Installing MINEOS, Identifying normal modes, Synthetic seismograms by summing normal modes	02	Earth's normal modes, computing synthetic seismograms.
10.	Estimating surface wave dispersion from ambient noise, Phase and group velocity dispersion from earthquake data.	03	Surface waves and its dispersion phenomena.
11.	Seismic tomography, Receiver function computation, H-κ stacking, CCP stacking	04	Crustal structure using seismic tomography and receiver functions.
12.	Velocity analysis, Processing and measuring SsPmP arrivals, Computer programs in Seismology, Ray Tracing using TauP	03	Ray tracing and techniques in seismology for finding seismic velocity structure.
13.	Revision of the entire course	02	Revision.
	Total	**42**	

Text books

1. Herrmann, R. B., 2013. Computer programs in seismology: An evolving tool for instruction and research, Seism. Res. Lettr. 84, 1081-1088, doi:10.1785/0220110096.

2. Notes on Seismological Data Analysis supplied by the course instructor.

Reference books

1. Shearer, P., 1999. Introduction to Seismology, Cambridge: Cambridge University

2. Lowrie, W., 2007. Fundamental of Geophysics, Cambridge: Cambridge University Press.

3. Gubins D., 1990. Seismology and Plate Tectonics, Cambridge University Press, 348pp.

4. Agustin, U., 2000. Principles of Seismology, Cambridge: Cambridge University